

ADHCRI Health Intelligence Dashboard: Data-Driven Insights for ASEAN Public Health Resilience

SDG 3

Good Health & Well-being

SDG 10

Reduced Inequalities

SDG 17

Partnerships for the Goals

Contributors

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Healthsync Analytics Team

Our Slogan: **Data Analytics Focus**

The Challenge

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ASEAN needs faster, clearer public health intelligence to support resilient decision-making.

Core problem

Fragmented health data, unequal health investment, infectious disease burden, and climate-related health risks make it difficult to prioritise action across ASEAN countries.

01

Health inequality
Different levels of mortality,
nutrition risk and life
expectancy.

02

Disease burden
TB, malaria and HIV indicators
require stronger monitoring.

03

System capacity
Workforce and expenditure vary
across countries.

04

Decision gaps
Policy leaders need simple
priority signals from data.

Data → Insight → Action

The dashboard is designed as a decision-support tool, not just a chart collection.

Main analytical question

Which ASEAN countries face the greatest public health risks, and where should policymakers prioritise investment to improve SDG 3 outcomes?

1

Identify health gaps

Compare country-level outcomes across mortality, disease, nutrition and life expectancy.

2

Understand risk drivers

Connect child health, disease burden and system capacity indicators.

3

Guide policy action

Translate evidence into investment and intervention priorities.

Project output: an interactive Power BI dashboard supported by a report and presentation.

Dataset Overview

Historical ASEAN public health indicators aligned to SDG 3, SDG 10 and SDG 17.

ASEAN countries	Indicators used	Valid records	Year coverage
10	16+	1,691	1990–2016

Indicator themes	
Population Dynamics Crude birth / death rates	Disease Burden TB, malaria, HIV and HIV/AIDS deaths
Health Outcomes Life expectancy	Nutrition & Food Security Undernourished population and underweight children
Child & Maternal Health Infant, under-5 and maternal mortality	System Capacity Health expenditure and workforce density

Important limitation: the data is historical and should be interpreted as SDG trend analysis, not a live 2026 health surveillance system.

Power Query & Data Model

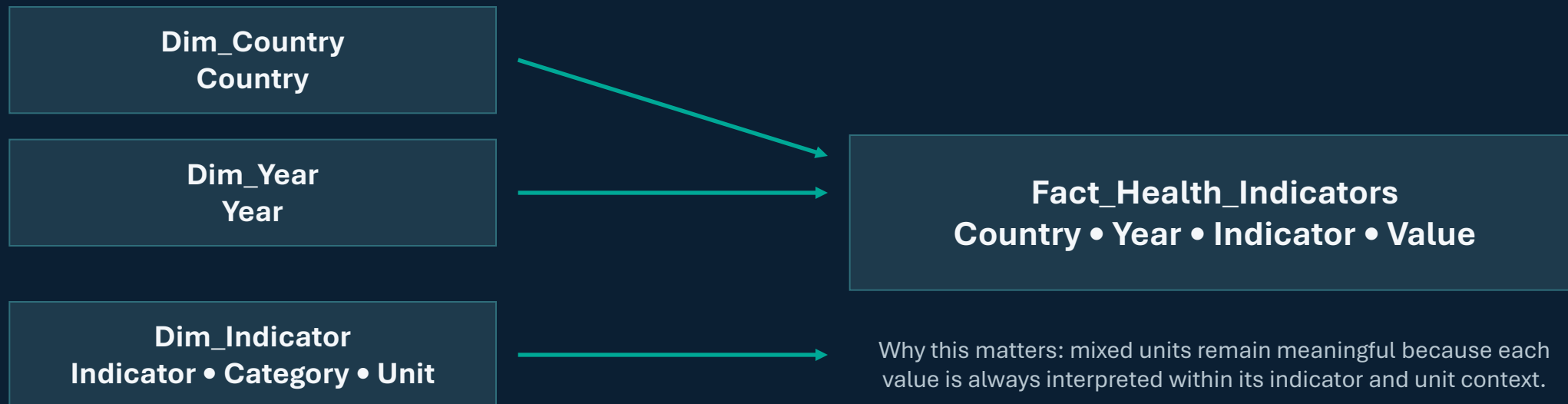
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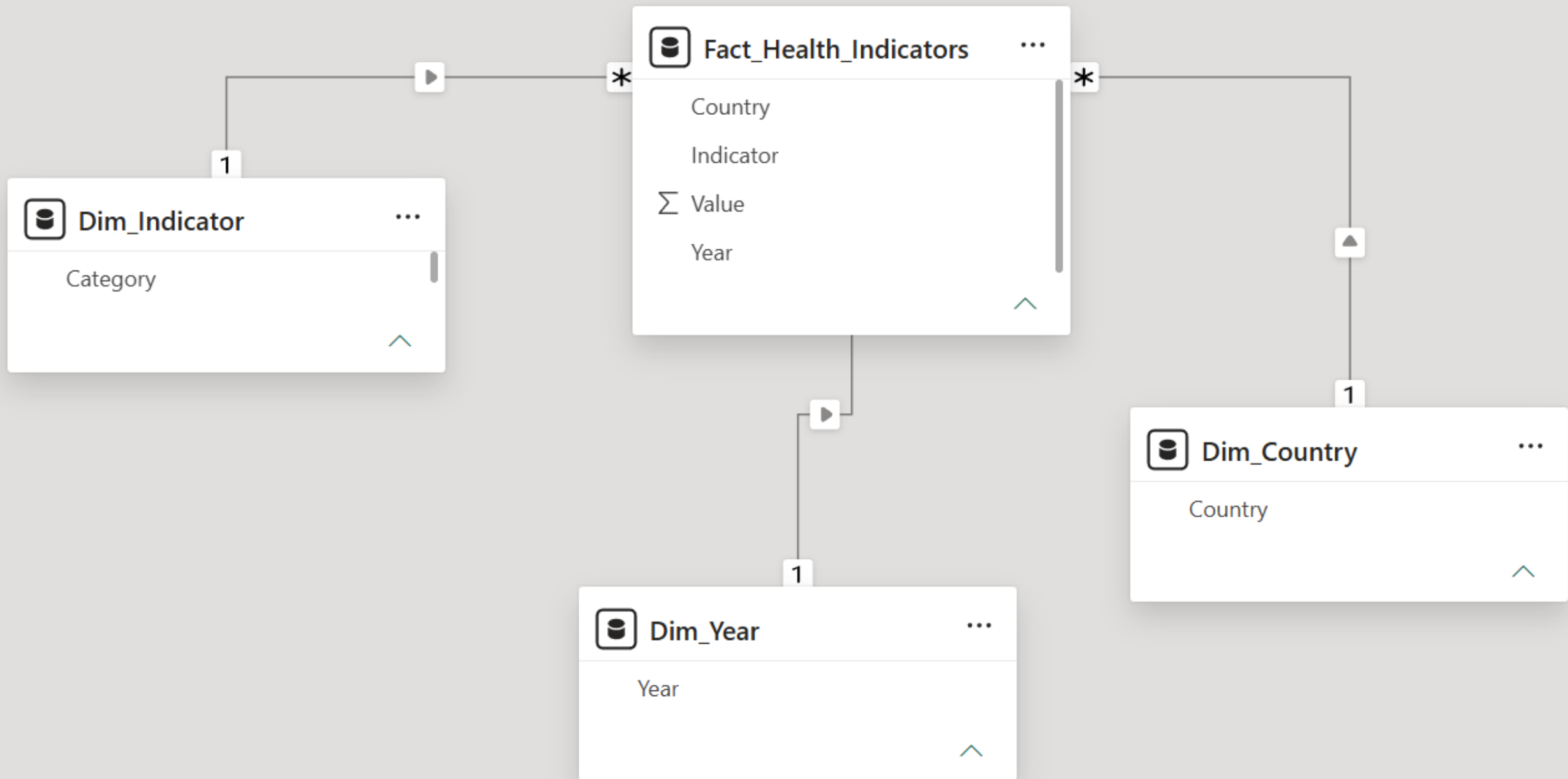
The raw files were standardised into a clean fact table and dimension model.

Cleaning workflow



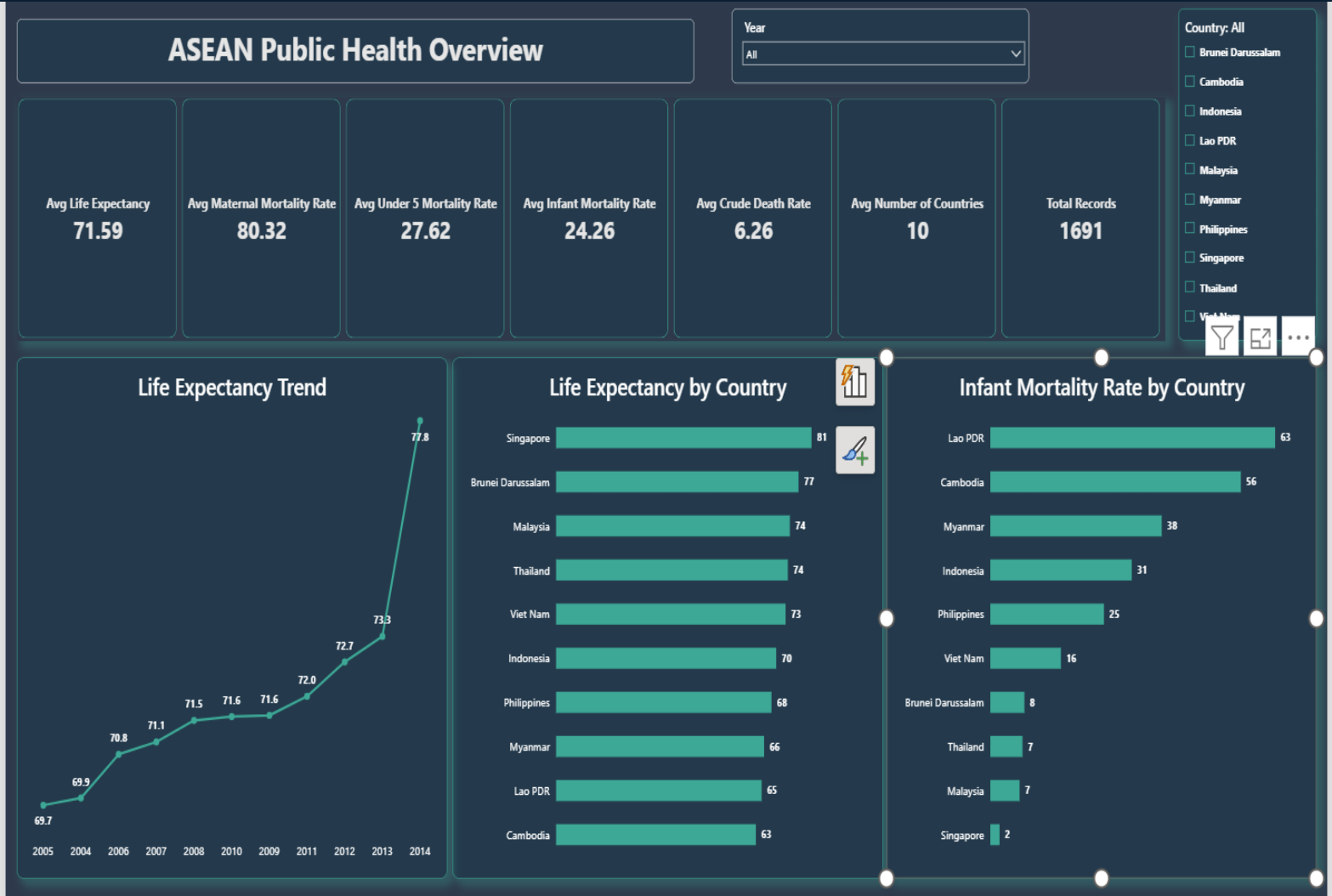
Star schema





Dashboard Page 1: Public Health Overview

A high-level view of ASEAN health outcomes and mortality gaps.



What this page shows

- Life expectancy differences
- Infant and under-5 mortality
- Maternal mortality
- Crude death rate
- Country and year filtering

Key insight

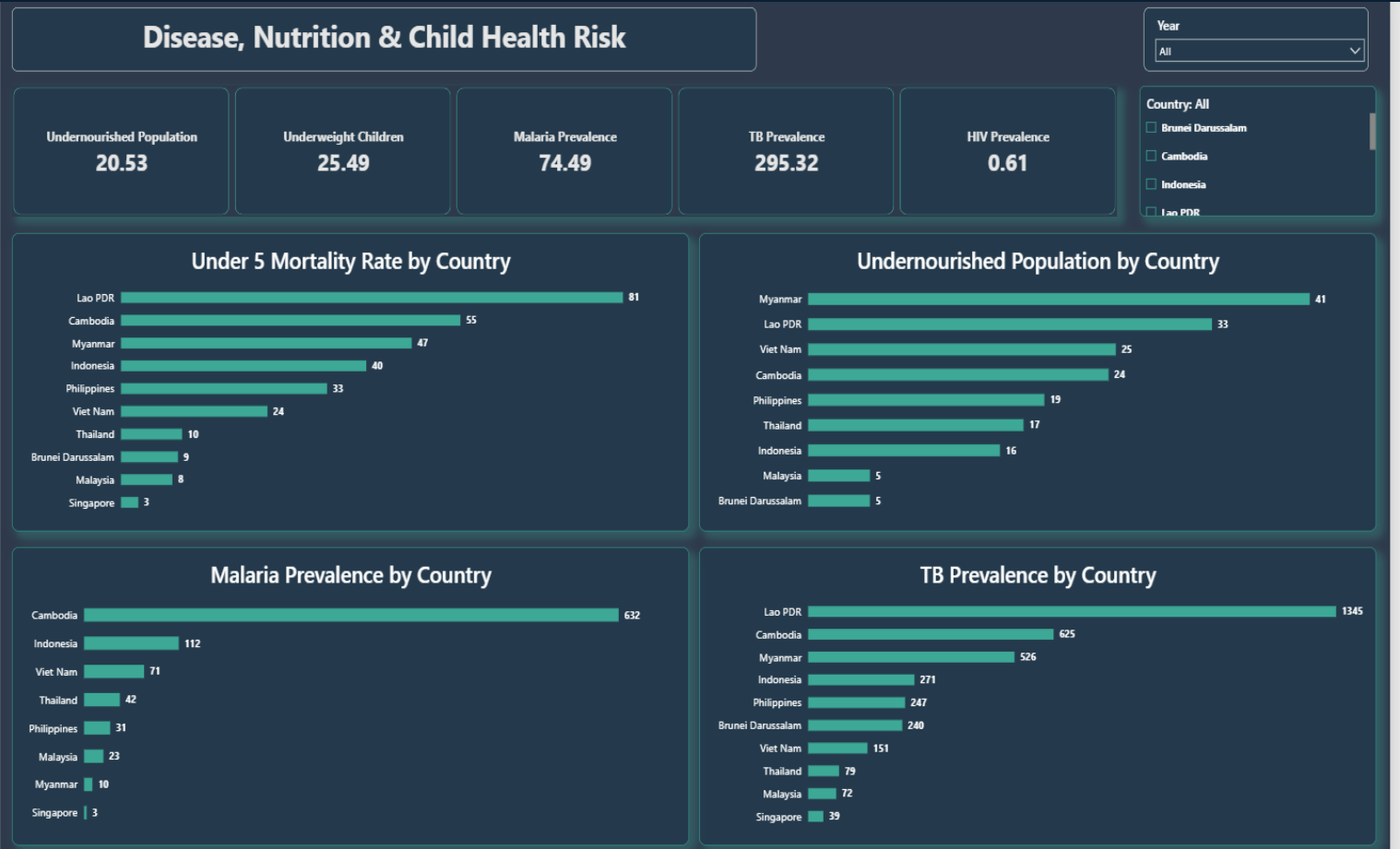
Life expectancy differs across ASEAN countries, while child and maternal mortality indicators reveal clear public health outcome gaps.

Design note

Rates and percentages are analysed as averages within each selected indicator to avoid mixing incompatible units.

Dashboard Page 2: Health Risk

Disease, nutrition and child health indicators identify higher vulnerability across ASEAN countries.



Healthcare Capacity & Investment

Avg Govt Health Exp
335.85

Avg Physicians Density
0.77

Avg Nurses Midwives Density
2.44

Avg Pharma Worker Density
0.19

Year

All

Country: All

Brunei Darussalam

Indonesia

Malaysia

Philippines

Thailand

Cambodia

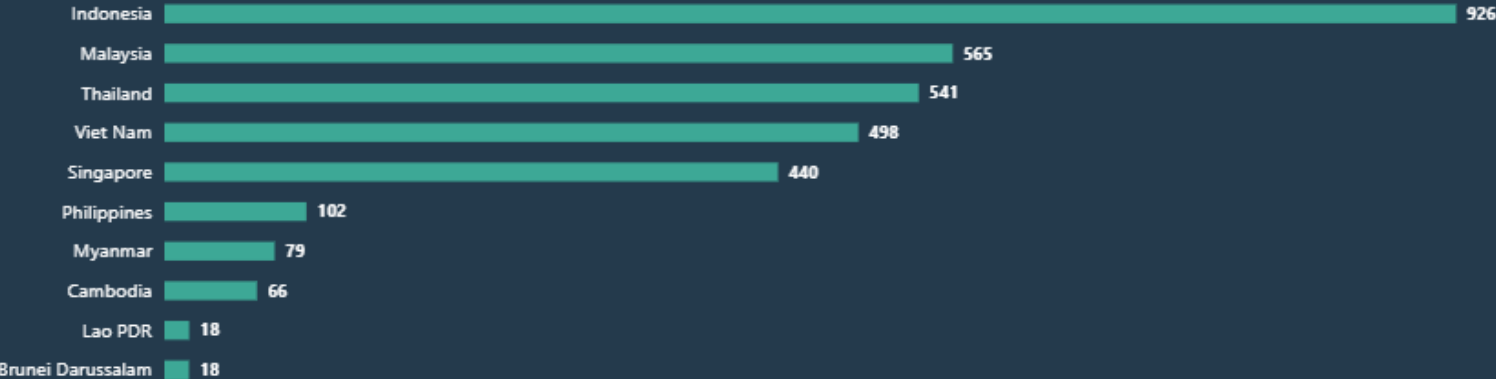
Lao PDR

Myanmar

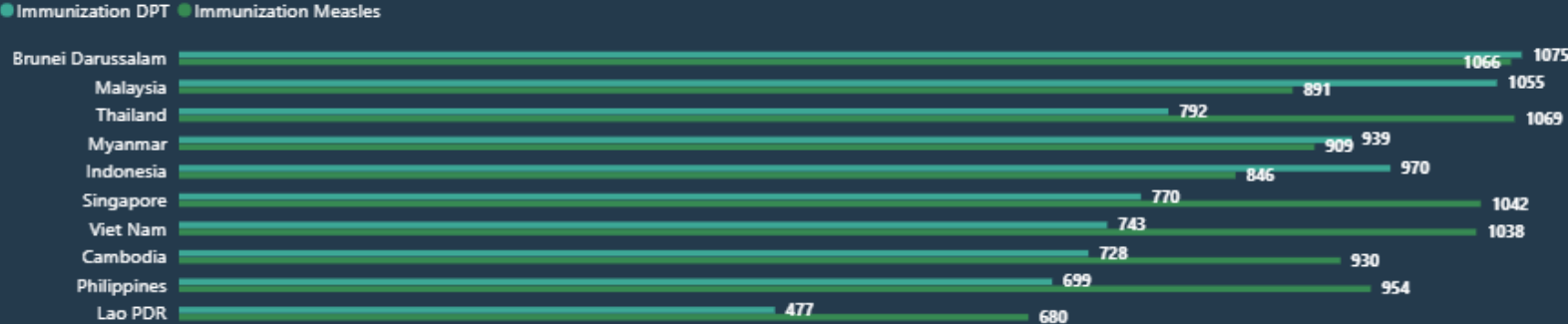
Singapore

Viet Nam

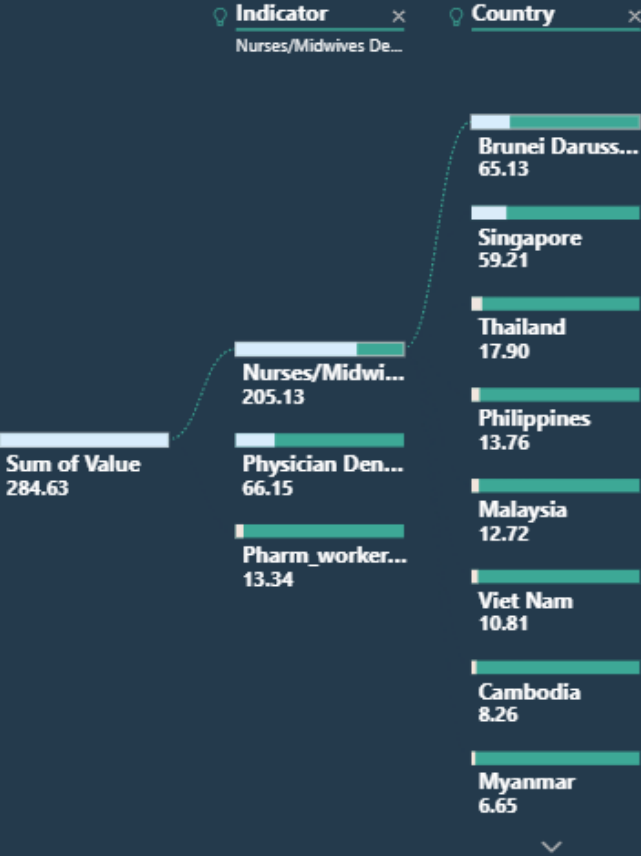
Government Health Expenditure by Country



DPT vs Measles Immunisation by Country



Healthcare Workforce



Dashboard Page 3: Capacity & Investment

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This page links health outcomes to healthcare financing, workforce and prevention capacity.

Government Health Expenditure
Investment capacity and infrastructure readiness

Physician Density
Medical workforce availability

Nurses/Midwives Density
Maternal, child and community care capacity

DPT & Measles Immunisation
Preventive healthcare coverage

- Which countries invest more or less in healthcare?
- Which countries have lower workforce density?
- How strong is immunisation coverage across countries?
- Where could limited capacity weaken public health response?

Emerging insight
Healthcare investment and workforce capacity vary across ASEAN countries. This may affect disease response, maternal care, child health support and access to essential services.

Key Findings

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The analysis points to unequal outcomes, disease burden and investment capacity across the region.

Funding gap

Indonesia recorded the highest government health expenditure, while Brunei Darussalam recorded the lowest in the analysed data.

Child health inequality

Lao PDR recorded the highest under-5 mortality rate, while Singapore recorded the lowest.

Disease burden concentration

TB and malaria prevalence are concentrated in selected countries, showing the need for targeted surveillance.

Life expectancy progress

Life expectancy improved across the analysed period, but country-level gaps remain visible.

Recommendations & SDG Impact

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Practical actions for policymakers and development partners.

Recommendations

- Prioritise child and maternal health investment in high-risk countries.
- Strengthen TB, malaria and HIV surveillance using routine data dashboards.
- Improve immunisation coverage and nutrition support for children under five.
- Increase healthcare workforce capacity where density is low.
- Use ASEAN-wide data collaboration to track SDG 3 progress consistently.

SDG alignment

SDG 3

Tracks mortality, disease burden, immunisation, nutrition and workforce indicators.

SDG 10

Highlights inequality in health outcomes and health system capacity.

SDG 17

Supports regional data partnerships and evidence-based public health planning.

ADHCRI Health Intelligence Dashboard

A 10Alytics Global Hackathon 2026 project using analytics to highlight health risks, inequalities and priority areas for action across ASEAN.

- Good Health & Well-being
- Reduced Inequalities
- Partnerships for the Goals

Thank You